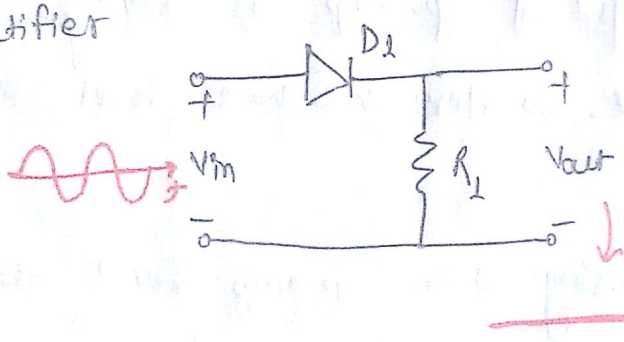
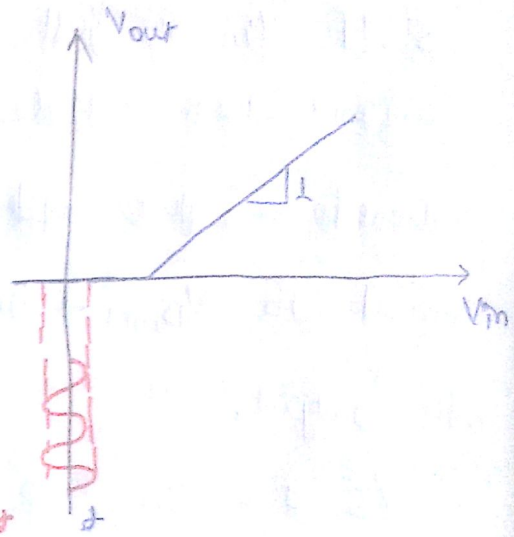
Lecture - 44

* Precision Rectifier :-

Simple rectifier





* If the sinusoid amplitude is smaller then the output is zero. As long as the amplitude of input sinusoid is less than $V_{D,ON}$, we did not output. This range is blind or dead zone for the input.

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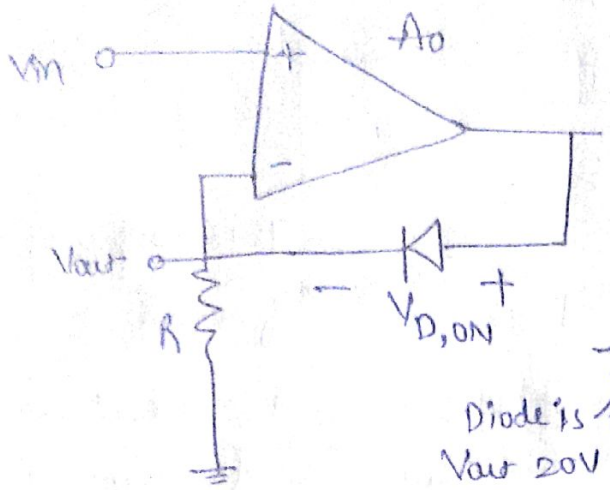

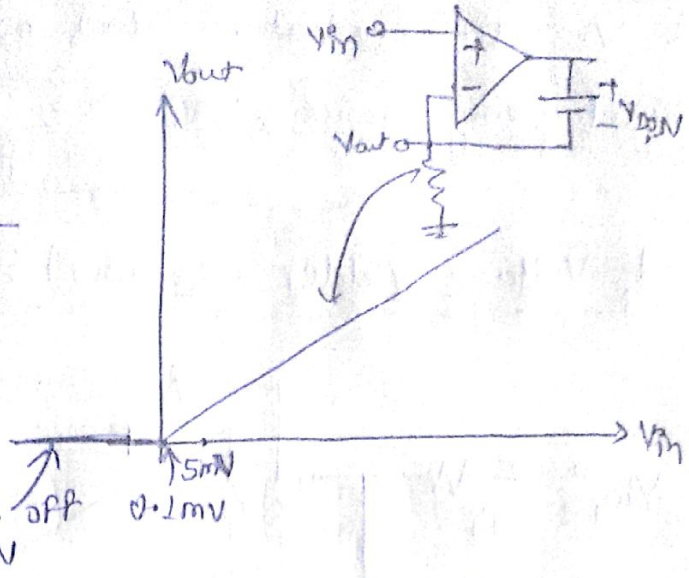

* When the input become little positive even input is 5mV the output goes to 5mV. As the ~~output~~ input becomes more positive the output follows it. As input becomes negative output becomes zero. So it like a ideal rectifier.

* If Ao is high →

→ If V_{in} is just a few mV (5mV) then we assume V_{out} is 0V and gain is high then we get 5V which is not possible because diode will turn ON and it wants to maintain a difference of about 700mV. So the V_{out} cannot be zero if V_{in} is 5mV.

→ If V_{in} is just 0.1mV, we assume that V_{out} is also about 0.1mV then diode becomes turn ON with a very small current. Therefore, the output of op-amp is sitting at the voltage equal to $V_{D,ON}$ + 0.1mV. So Here we have level shift between both output.

As the V_{in} keep increasing then op-amp output also rises except that this voltage is always $V_{D,ON}$ above V_{out} or V_{in}. Therefore V_{out} follows V_{in}.

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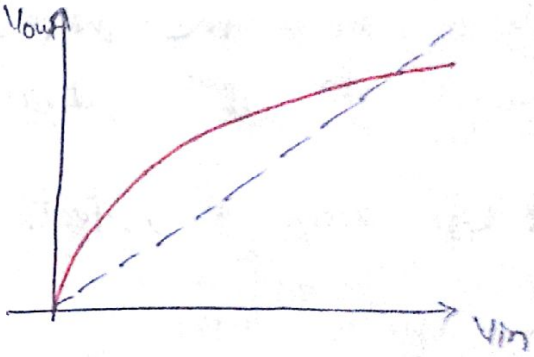
→ If V_{in} is negative so the V_{out} cannot be negative because diode cannot conduct current in opposite direction.

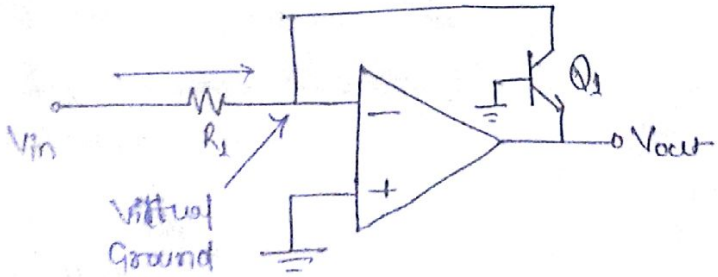
* If we have sinusoidal waveform to determine its amplitude, first we rectify it then we find the average means Low pass filter it.

* Precision Rectifier is used to rectify even small signals.

Logarithmic Amplifier:-

It amplify smaller signals with larger gain and large signals with smaller gain.





$$\text{Current through } R_1 \approx \frac{V_{in}}{R_1} = I_{C_1} = I_S \exp\left(\frac{V_{BE}}{V_T}\right)$$

$$V_{BE} = V_B - V_E = 0 - V_{out} = -V_{out}$$

$$\Rightarrow \frac{V_{in}}{R_1} \approx I_S \exp\left(\frac{-V_{out}}{V_T}\right) \Rightarrow \boxed{V_{out} = -V_T \ln \frac{V_{in}}{I_S R_1}}$$

If V_{in} is -ve so transistor turns off because current cannot flow from emitter to collector in npn transistor. Output of op-amp is not defined (saturated).

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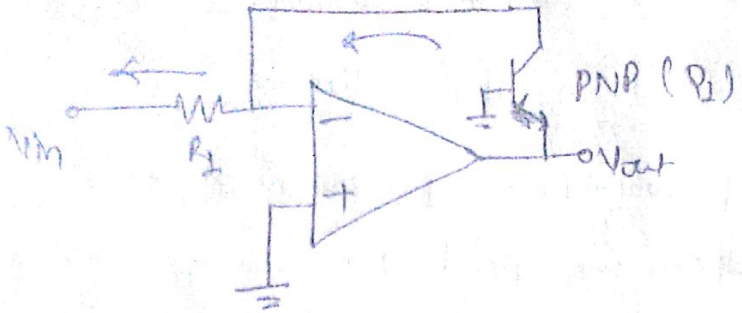
Quiz:



$V_{BE} = -V_{out}$

$$I_C \simeq -\frac{V_{in}}{R_1} = I_S \exp\left(\frac{-V_{BE}}{V_T}\right) = I_S \exp\left(\frac{+V_{out}}{V_T}\right)$$

$$\boxed{V_{out} = +V_T \ln\left(\frac{-V_{in}}{R_1 I_S}\right)}$$

When V_{in} is +ve then transistor cutoff because it cannot conduct current into opposite direction.

---

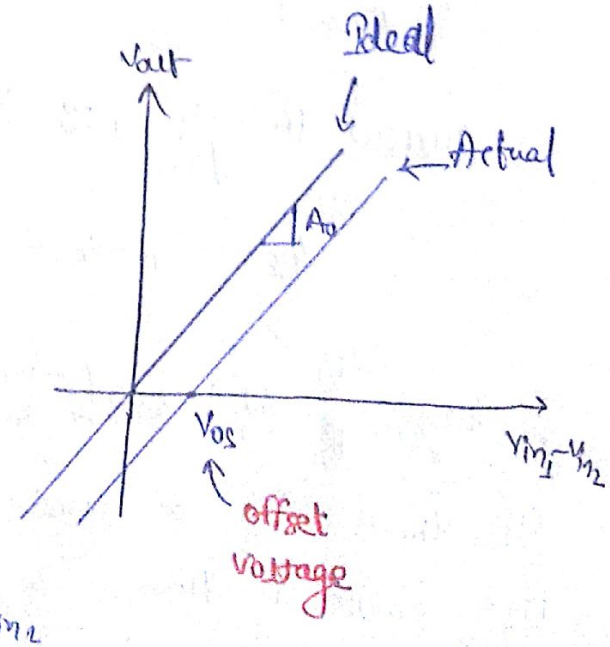
* Op-amp Imperfections/non idealities :-

- Finite gain
- DC offsets
- Input bias currents
- Finite speed

* DC offsets :-

→ In actual the V_{out} v/s $(V_{in_1} - V_{in_2})$ is not same as ideal case.

→ If $V_{in_1} = V_{in_2}$, output is not zero alternatively, we can say the output to be zero we have to change $V_{in_1} - V_{in_2}$ to new value V_{os}. To force V_{out} to be zero, we add something between V_{in_1} & V_{in_2} to bring it at 0, this voltage is called offset voltage.



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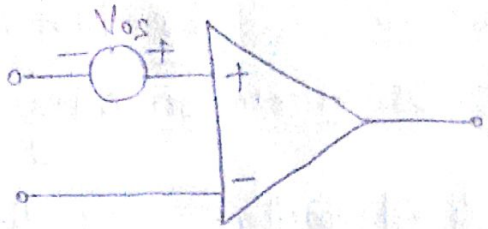
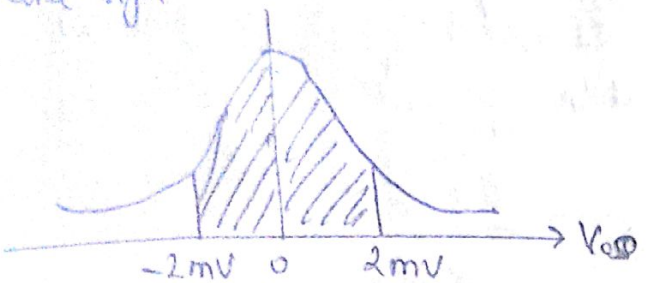
Observations:-



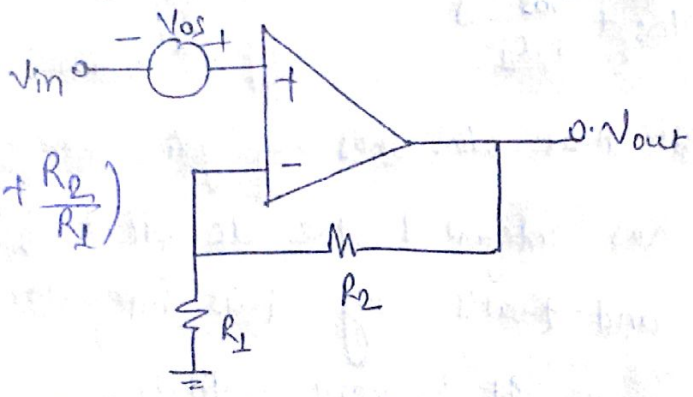
① The V_{os} has random value and sign.



② V_{os} can be placed in series with either input.

Effect of offsets on Amplifiers:-

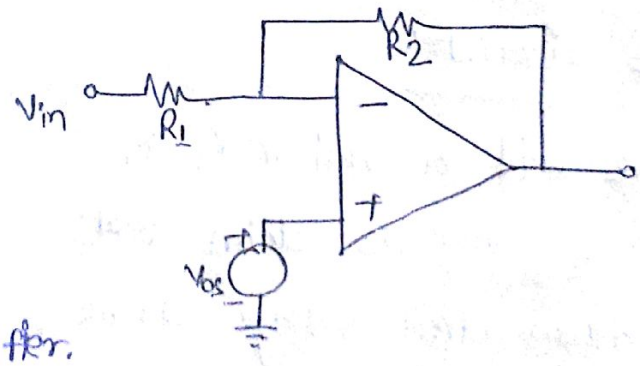
①



$$V_{out} = (V_{in} + V_{os})\left(1 + \frac{R_2}{R_1}\right)$$

In this offset is also amplified which can corrupt the signals.

② In this, by superposition when V_{os} is zero it is like inverting amplifier and when V_{in} is zero it is like non inverting amplifier.



$$V_{out} = -\frac{R_2}{R_1} V_{in} + V_{os}\left(1 + \frac{R_2}{R_1}\right)$$

Since offset voltage is amplified by factor greater than the factor by which input signal is amplified, therefore it corrupts the signals.

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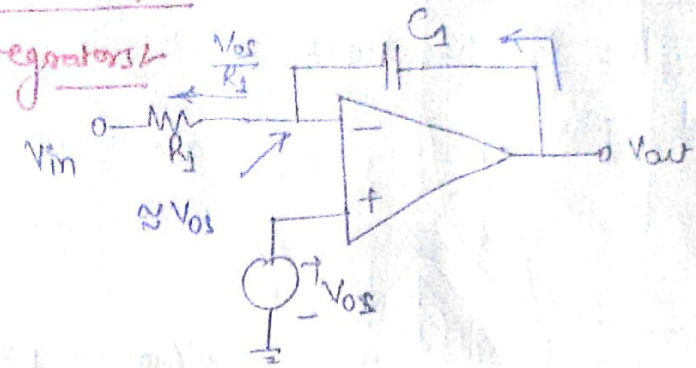
Lecture - 45

* Effect of DC offsets on Integrators:-



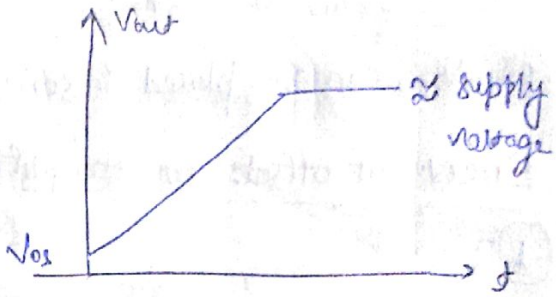
Current through an capacitor

$$I = C\frac{dV}{dt}$$

$$\Rightarrow V = \frac{1}{C_1}\int I\,dt$$

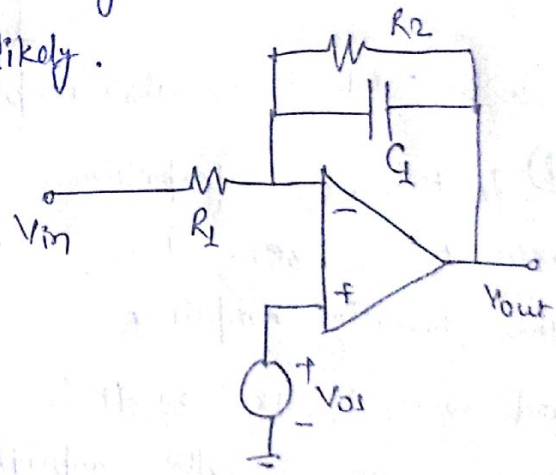
$$= \frac{1}{C_1}\int \frac{V_{os}}{R_1}dt = \frac{1}{R_1C_1}\int V_{os}\,dt$$

$$\Rightarrow V_{out} = V_{os} + \frac{V_{os}}{R_1C_1}t$$



* The integrator at that point has no gain therefore it becomes dead.
* The integrator has integrated the dc offset of the op-amp and reached saturation and that's why this integrator does not work in the lab because it is very unlikely.

To fix the circuit:-



* In this we add a resistor R_2 in parallel with C_1 and we claim that R_2 does not allow offset voltage to be integrated.

If V_{in} is zero,

$$V_{out} = \left(1 + \frac{R_2}{R_1}\right)V_{os}$$

So we have some error in this.

$\longrightarrow$ $\frac{V_{os}}{R_1}$ prefers to flow through R_2 rather than through C_1 at $t = \infty$.

Ideal Integrator Transfer function: $-\frac{1}{R_1C_1s}$

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New Integrator:- $\frac{V_{out}}{V_{in}} = -\frac{Z_2}{Z_1} = -\frac{\frac{R_2}{R_2C_1s+1}}{R_1} = -\frac{R_2}{R_1(R_2C_1s+1)}$

For to be a good integrator, the circuit requires that

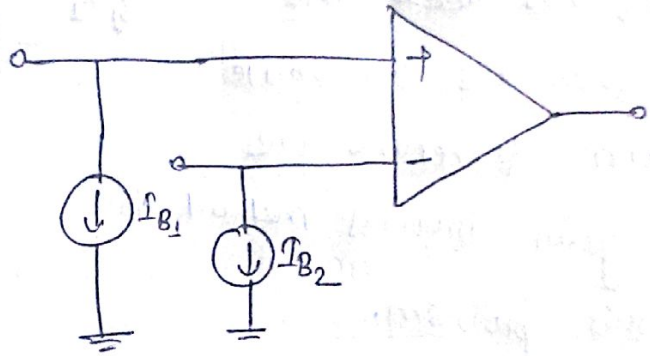
$$|R_2C_1s| \gg 1$$

For frequency to be large, it is good integrator and for frequency to be small, it is not a good integrator.

---

* Input Bias currents:- If we buy a bipolar opamp (741) then we do need to remember that input needs some bias currents. If the device does not have bias currents then it is dead.

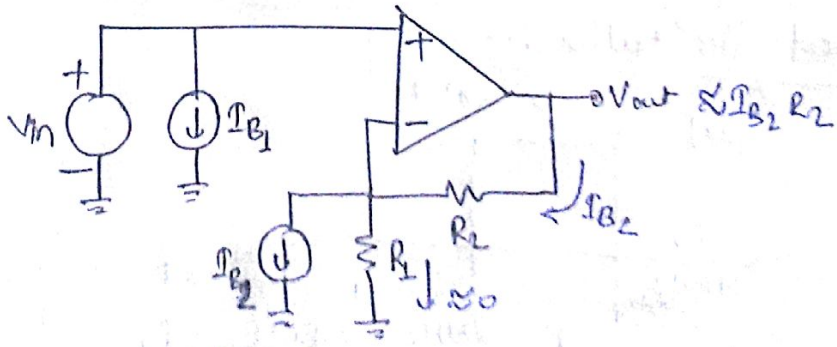
* If we have CMOS op-amp then we do not have that problem.

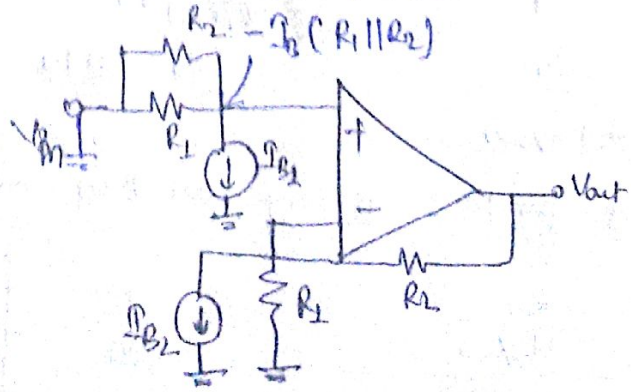


① Say $V_{in} = 0$

$V_{out} \approx R_2 I_{B_2}$

To cancel it we have to substract this value some how.





$$V_{out} = -I_{B_1}(R_1 \| R_2)\left(1 + \frac{R_2}{R_1}\right) + I_{B_2}R_2$$

$$= I_{B_2}R_2 - I_{B_1}R_2$$

$$= 0$$

Therefore we dont have any dc value because of input bias currents.

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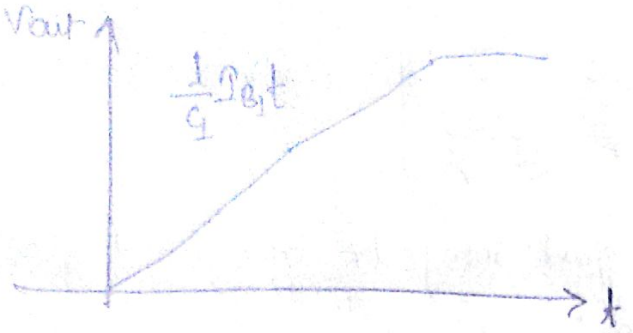
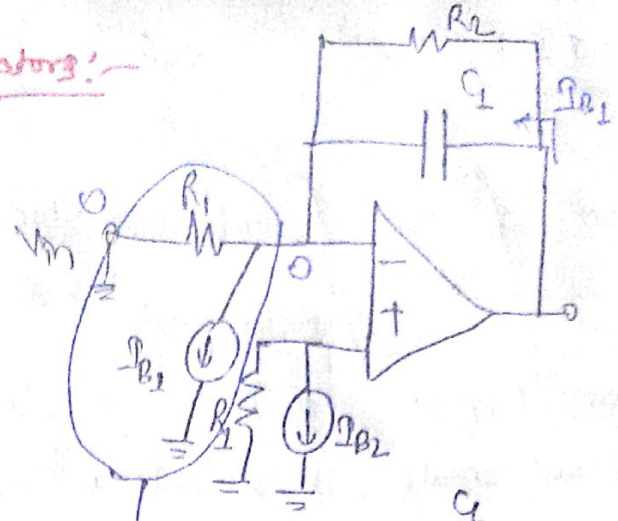
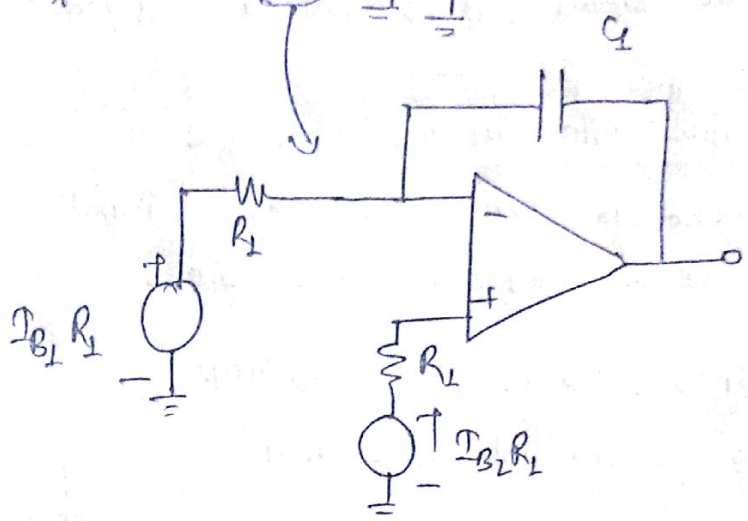
Input bias currents in Integrators:-

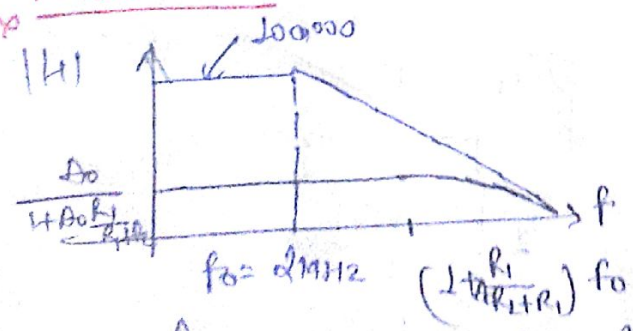


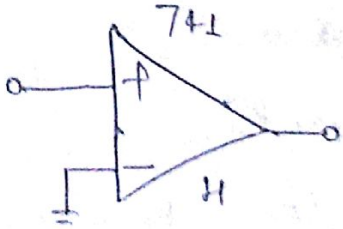




We use this method to solve this problem but if $I_{B_1} - I_{B_2}$ has some value then this can create problem therefore we use again previous method to solve this problem.

Speed limitations:-



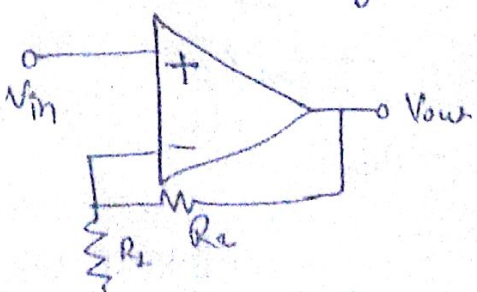


$$H(s) = \frac{A_0}{1+s/\omega_0}$$

$\omega_0 = 2\pi f_0$

If frequency of operations is very low then we have only A_0.

$$\frac{V_{out}}{V_{in}}(s) = \frac{\frac{A_0}{1+s/\omega_0}}{1+\frac{A_0}{1+s/\omega_0}\left(\frac{R_1}{R_1+R_2}\right)}$$



when we assume gain is frequency dependent.

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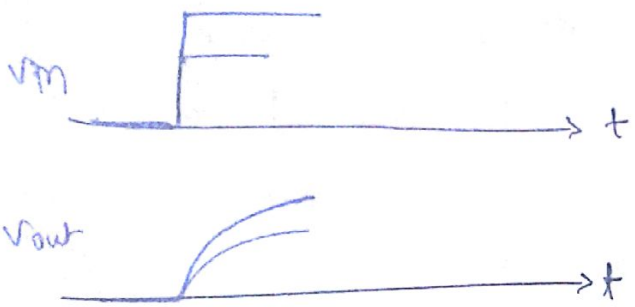
$$\frac{V_{out}}{V_{in}}(s) = \frac{A_0}{\frac{s}{\omega_0} + 1 + A_0\left(\frac{R_1}{R_1+R_2}\right)}$$

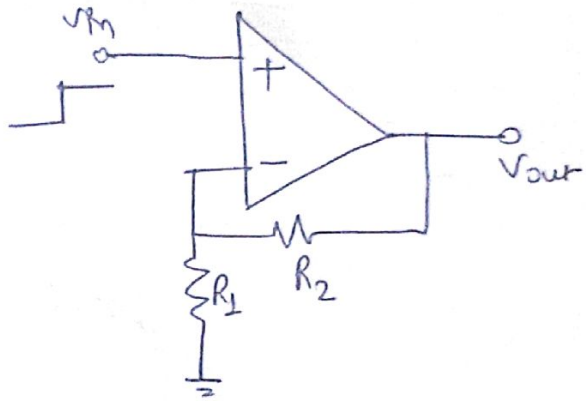
If s/ω_0 is low then gain is lower than the op-amp itself.

$$\text{pole freq}^n = -\left(1 + A_0\frac{R_1}{R_1+R_2}\right)\omega_0$$

we reduced the gain of this ckt from 100000 to that value but gain when down bandwidth increases from 2MHz to a huge amount.

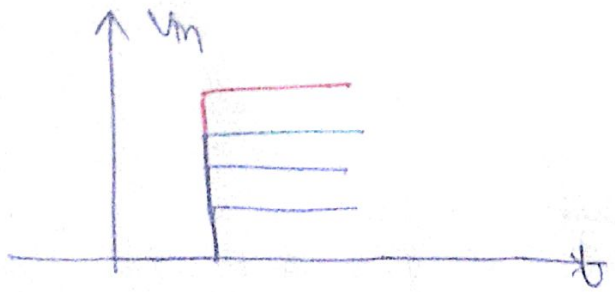
Slew Rate:-

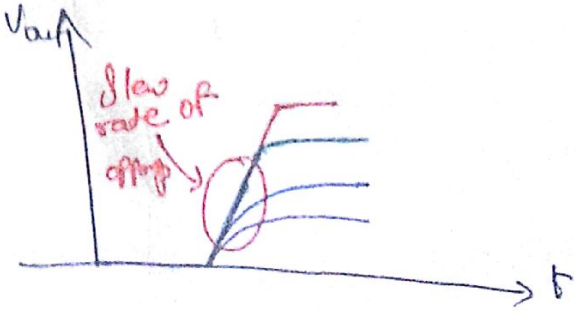




If we apply step at Vin then we see an exponential at Vout. If we increase height of step then the exponential at output has slope also increases.

If we perform this on real op-amp then,





This slope fails to scale like before if we height of slope is high. This slope becomes constant it does not increase. This slope is called slew rate of op amp. This ~~process~~ is called slewing effect.

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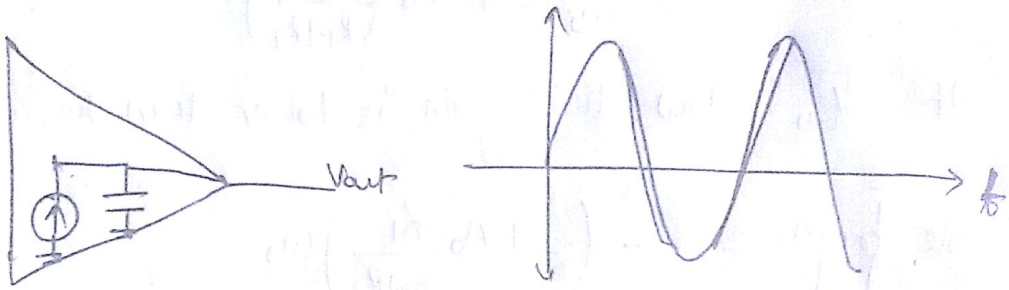
* In this case the op amp looks like constant current charging a capacitor.



* This effect limits the speed of op-amp.

* We have original waveform like sinusoid which get distorted and go as a ramp.

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